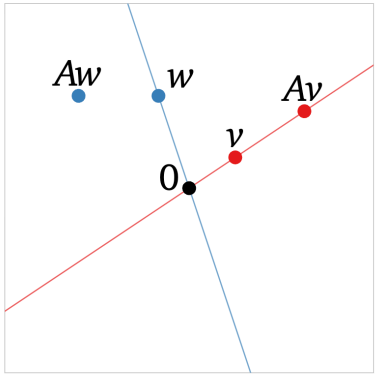
Eigenvalues and Eigenvectors

Definition. Let A be an $n \times n$ matrix.

1. An ***eigenvector*** of A is a *nonzero* vector v in $\mathbf{R}^n$ such that $Av = \lambda v$, for some scalar λ.
2. An ***eigenvalue*** of A is a scalar λ such that the equation $Av = \lambda v$ has a *nontrivial* solution.

If $Av = \lambda v$ for $v \neq 0$, we say that λ is the ***eigenvalue for*** v, and that v is an ***eigenvector for*** λ.

Geometrically:



v is an eigenvector

w is not an eigenvector

Recipes: Eigenspaces.

Let A be an $n \times n$ matrix and let λ be a number.

1. λ is an eigenvalue of A if and only if $(A - \lambda I_n)v = 0$ has a nontrivial solution, if and only if $\text{Nul}(A - \lambda I_n) \neq \{0\}$.
2. In this case, finding a basis for the λ-eigenspace of A means finding a basis for $\text{Nul}(A - \lambda I_n)$, which can be done by finding the parametric vector form of the solutions of the homogeneous system of equations $(A - \lambda I_n)v = 0$.
3. The dimension of the λ-eigenspace of A is equal to the number of free variables in the system of equations $(A - \lambda I_n)v = 0$, which is the number of columns of $A - \lambda I_n$ without pivots.
4. The eigenvectors with eigenvalue λ are the nonzero vectors in $\text{Nul}(A - \lambda I_n)$, or equivalently, the nontrivial solutions of $(A - \lambda I_n)v = 0$.

Definition. Let A be an $n \times n$ matrix, and let λ be an eigenvalue of A. The ***λ-eigenspace*** of A is the solution set of $(A - \lambda I_n)v = 0$, i.e., the subspace $\text{Nul}(A - \lambda I_n)$.

eigenvector corresponding to eigenvalue λ cannot be $\vec{0}$

λ-eigenspace must contain $\vec{0}$: $\{\vec{v} \mid A\vec{v} = \lambda\vec{v}\} = \{\vec{v} \mid (A - \lambda I_n)\vec{v} = 0\}$.